

- 2 The photograph shows a rabbit with brown fur from Lokrum Island near Croatia.



(Source: © Peter Shaw / Alamy Stock Photo)

- (a) The Hardy-Weinberg equation can be used to determine changes in allele frequencies over time.

The equation can be written as $p^2 + 2pq + q^2 = 1$

Rabbits that inherit a dominant B allele have brown fur.

The proportion of homozygous dominant (BB) rabbits in the island population is represented by p^2 .

- (i) Which is the value for $2pq$ if $p^2 = 0.36$?

(1)

- ☐ **A** 0.06
- ☐ **B** 0.16
- ☐ **C** 0.48
- ☐ **D** 0.64

- (ii) Albino rabbits in this population have the genotype bb.

State the part of the equation that represents this genotype.

(1)



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(b) A viral disease has killed many of the rabbits on Lokrum Island.

Several genes are involved in protecting rabbits from this virus.

Explain how this viral disease could result in a change in allele frequencies of **these** genes in the rabbit population on Lokrum Island.

(4)

(Total for Question 2 = 6 marks)

